

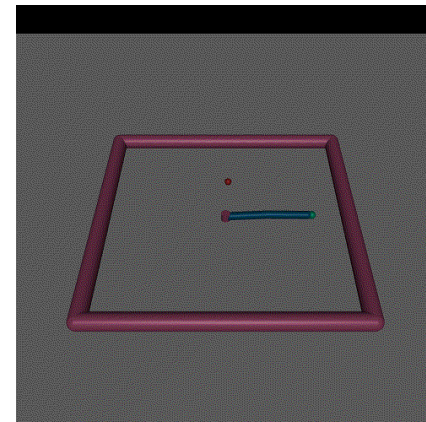
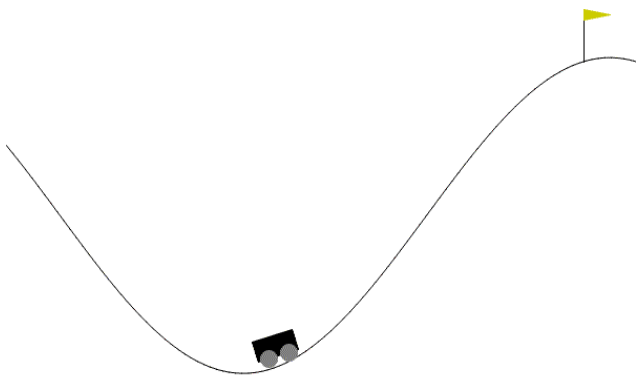
# Deep Q Learning

An Application on Grid World

Jan 06, 2025, Jiang YOU

# Introduction

- Reinforcement learning
  - Reinforcement learning is a subfield of machine learning where an agent learns to make decisions in an environment to maximize a cumulative reward signal.
  - Models: Qlearning, Sarsa, DQN, DDPG...
- Examples of applications of reinforcement learning in various fields
  - Robotics, Game, Autonomous vehicles, Finance, Healthcare, Advertising
  - OpenAI gym : <https://www.gymnasium.dev/>



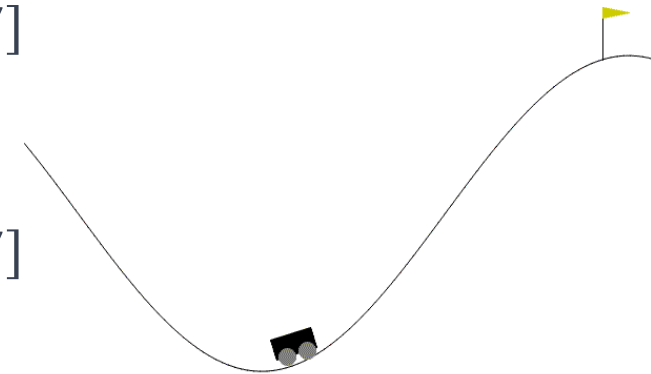
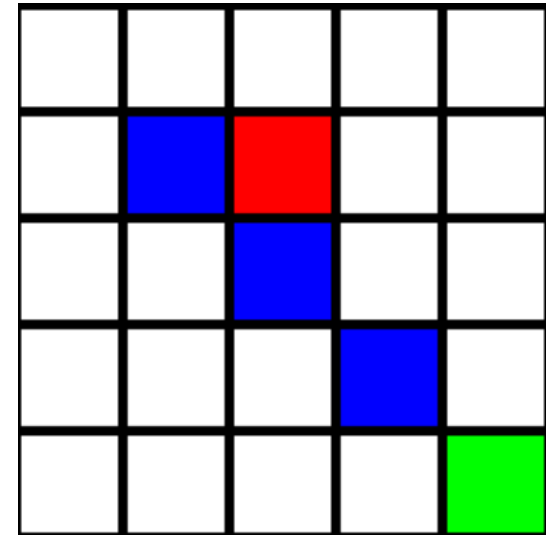
# Q-learning and Sarsa

- Qlearning and Sarsa are model-free reinforcement learning algorithms that learn to estimate the value of an action in a particular state
- The difference of Qlearning and Sarsa is that
  - Q-learning is greedy and it learns from the action with a maximal q value in its current policy.
    - $Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha (r_t + \gamma \text{Max}_{a'} Q(s_{t+1}, a') - Q(s_t, a_t))$
  - Sarsa learns also from random actions when epsilon is smaller than a threshold.
    - $Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha (r_t + \gamma Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t))"$

# Reinforcement learning

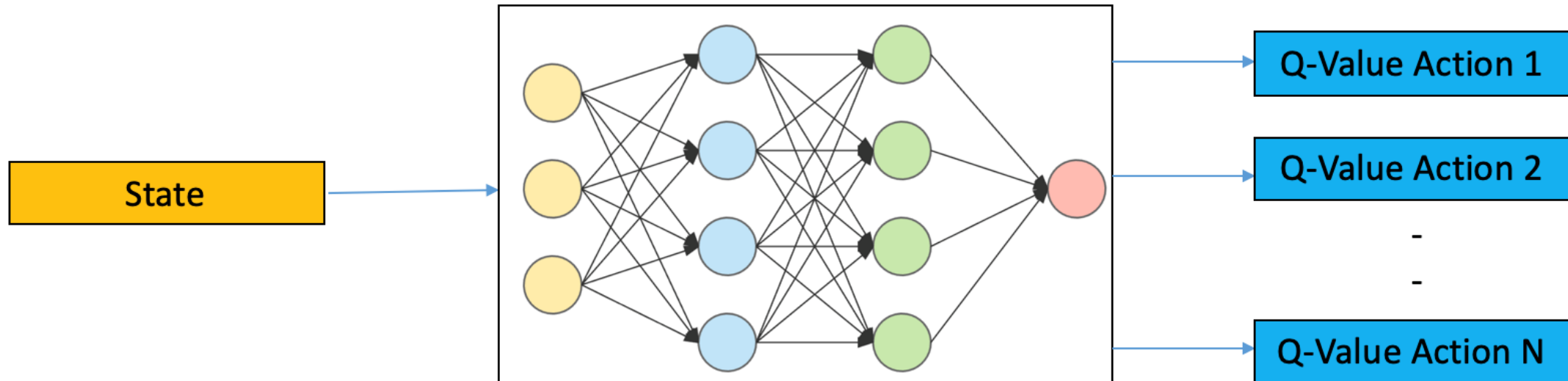
Reinforcement learning have many limitations

- The curse of dimensionality :
  - Grid world
    - State size:  $5 \times 5 = 25$  (Player position )
    - Action size : 4
  - Mountain car:
    - State size: position  $p \in [-1.2, 0.6]$ , and velocity  $v \in [-0.07, 0.07]$
    - Action size :  $a \in [\text{Left}, \text{None}, \text{Right}]$
  - Mountain car Continuous:
    - State size: position  $p \in [-1.2, 0.6]$ , and velocity  $v \in [-0.07, 0.07]$
    - Action size : acceleration  $a \in [-1, 1]$



# Deep Q-learning

Deep Q-learning agent is an extension of the Q-learning algorithm that uses a deep neural network to estimate the Q-values instead of a Q-table. This makes it possible to handle high-dimensional state spaces and large action spaces more efficiently.

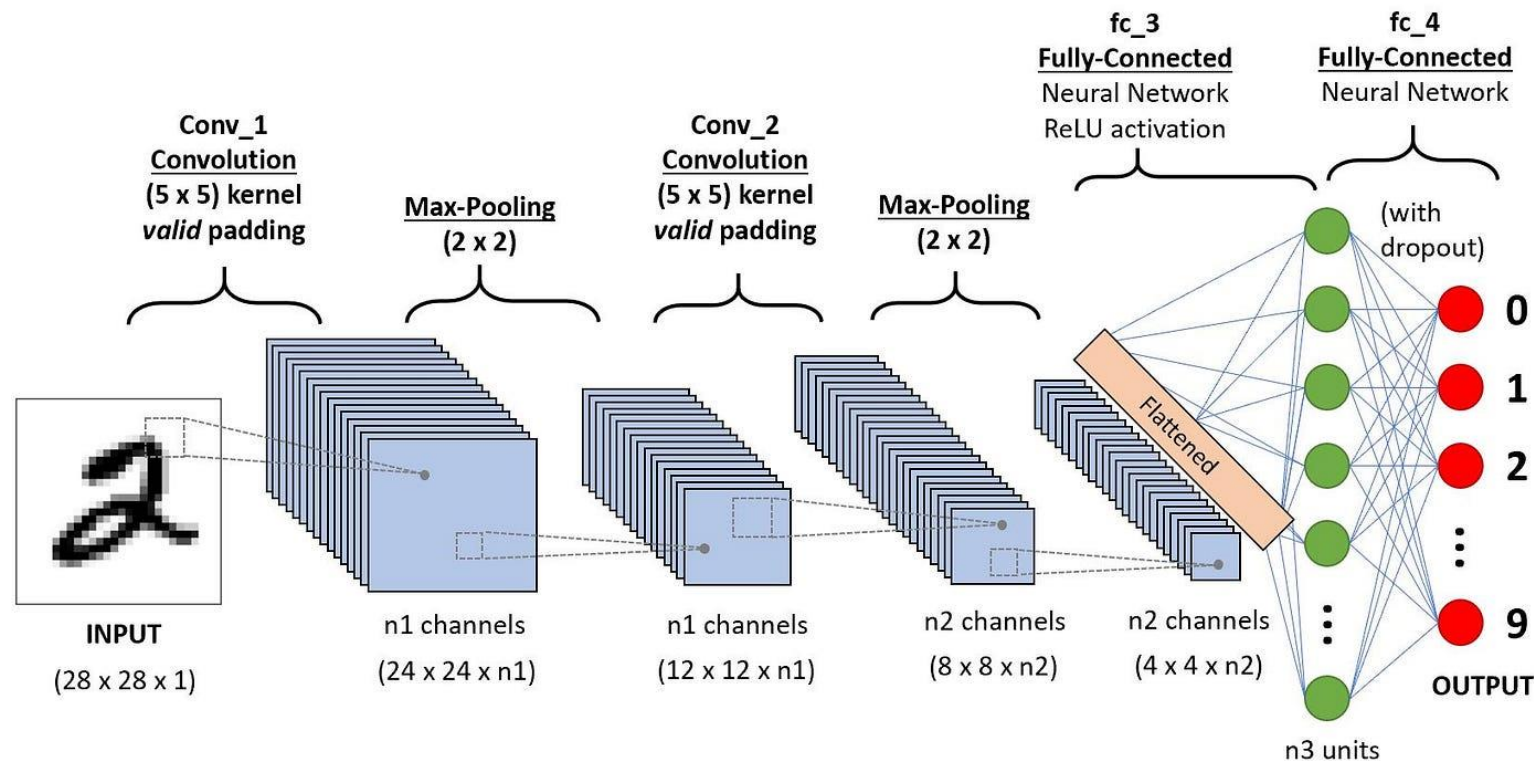


## Deep Q Learning

# Convolutional Neural Network

Convolutional Neural Networks (CNNs) are a type of neural network commonly used in image recognition and computer vision tasks.

They are designed to automatically learn and extract features from images through convolutional layers that apply filters to the input image.



# Replay Buffer Memory

- A replay buffer is stores past experiences (i.e., state-action-reward-next state tuples) of the agent in a memory buffer to be randomly sampled during the learning process.
- By replaying past experiences, the agent can learn from a more diverse and less correlated set of experiences, which can lead to more stable and efficient learning.
- Optimization:
  - Positive reward memory, Negative reward memory, trajectory memory
  - Prioritized Experience Replay Memory

# Loss function in DQN

- The loss function in Deep Qlearning is the temporal difference error between target Q value and current Q value
  - The Bellman equation is the same to Q-learning.
    - $Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha (r_t + \gamma \text{Max}_{a'} Q(s_{t+1}, a') - Q(s_t, a_t))$
  - The loss function used for neural network is defined as :
    - $\text{Min}_{\theta} \|Q_{\text{target}}(s_t, a_t) - Q(s_t, a_t)\| = \text{Min}_{\theta} \|r + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t)\|$

# Deep Q learning Agent

## Learning a DQN Agent

- Deep Q-Learning (DQN) agent first observes the current state of the environment.
- Then the agent selects an action based on its current Q-values, which are estimated using a neural network.
- The agent then executes the selected action in the environment, receives a reward signal and observes the next state.
- The agent stores this experience (state-action-reward-next state) in a replay buffer and samples a batch of experiences from the buffer to update the Q-values of the neural network using the Bellman equation and the Q-Learning loss function.

This process of sampling and updating the Q-values is repeated multiple times during training to improve the agent's performance over time.

